

EDUCATION

University of California, Irvine

Bachelor of Science in Physics; GPA: 3.83

Sep. 2023 - June 2027 (Expected)

Irvine, CA

Relevant Coursework:

- Quantum Mechanics I, II
- Electromagnetic Theory I, II
- Classical Mechanics
- Graduate Quantum Field Theory
- Graduate Quantum Mechanics
- Graduate Machine Learning and Statistics
- Graduate Group Theory
- Graduate Elementary Particle Physics

RESEARCH EXPERIENCE

Caltech SURF in Muon-Beam Dark Matter Searches

June 2026 - Aug. 2026

Mentor: Prof. Bertrand Echenard & David Hitlin, California Institute of Technology

- Implemented a detailed Geant4 simulation of a 40-layer LYSO crystal active target with per-crystal energy-deposition readout for an LDMX-style muon-beam missing-momentum dark matter search, and ran the full analysis chain at the 10^8 -muon scale on a Slurm cluster.
- Simulated and characterized the rare backgrounds mimicking the signal — muon bremsstrahlung with photonuclear interactions and hard muon–nucleus scattering — tracing each surviving event topology to its physical mechanism.
- Combined calorimetric veto thresholds with BDT, graph-neural-network, and transformer classifiers, reaching $\sim 90\%$ signal efficiency with zero surviving background events in the 10^8 -muon simulated exposure, under a preregistered statistical-testing protocol.
- Sole-authored a PRD-format manuscript covering the detector design, background mechanisms, and veto optimization.

Research on the Future Circular Collider (FCC-ee)

Jan. 2026 - Present

Mentor: Prof. Christoph Paus, Massachusetts Institute of Technology

- Built an end-to-end sensitivity study for $\sigma(\text{ZH}) \times \mathcal{B}(\text{H} \rightarrow \text{WW}^*)$ in the semileptonic $q\bar{q} \ell\nu q\bar{q}'$ final state at $\sqrt{s} = 240$ GeV, processing over 10^9 fully simulated events across 32 signal and background processes with the FCCAnalyses C++/RDataFrame framework.
- Designed a two-category measurement separating on-shell and off-shell leptonic- W decay topologies by the reconstructed $\ell\nu$ invariant mass (95–98% category purity, validated against generator truth), with an independently trained XGBoost classifier per category (test AUC 0.99).
- Implemented profile-likelihood inference in pyhf: a two-channel template fit projecting $\delta\mu/\mu = 0.55\%$ on the inclusive signal strength and separate per-topology strengths of 0.67%/0.91%, validated with Asimov closure, profile scans, and systematic cut-edge scans.
- Sole-authored a PRD-format manuscript and research talk; developed the Z-priority jet-pairing strategy and a prescription-based selection design validated against the full likelihood fit.

Phenomenology and ML of New Physics at the LHC

May 2025 - May 2026

Mentor: Prof. Daniel Whiteson, University of California, Irvine

- Led the development of a time-dependent resonance search strategy for new physics at the LHC, introducing event time as an auxiliary feature to improve sensitivity beyond mass-only searches.
- Extended CATHODE-based anomaly detection to model background distributions in invariant mass and time sidebands, enabling data-driven searches for temporally modulated signals.
- Simulated signal and background processes with MadGraph5, Pythia8, and Delphes, and built ML pipelines to compare mass-only and mass-plus-time search strategies.
- Quantified sensitivity gains using classifier performance, sideband closure diagnostics, and expected upper limits on signal strength.

Photometric Analysis of High-Redshift Galaxy Evolution

Dec. 2024 - Sep. 2025

Mentor: Prof. Asantha Cooray, University of California, Irvine

- Analyzed photometric data from Spitzer and ALMA telescopes to investigate the redshift distribution and physical properties of Herschel-selected ultra-red dusty star-forming galaxies.
- Developed automated pipelines in Python and CARTA to visualize spectral energy distributions (SEDs) and interpret high-redshift signals.

Nanophotonic Fabrication of Epsilon-Near-Zero Materials

Feb. 2024 - Sep. 2025

Mentor: Prof. Howard Lee, University of California, Irvine

- Fabricated gate-tunable 2D Indium Tin Oxide (ITO) films using liquid metal printing techniques for Epsilon-Near-Zero (ENZ) applications.
- Characterized optical nonlinearity and tunability of metasurfaces, contributing to the development of next-generation nanophotonic devices.

PUBLICATIONS & PRESENTATIONS

SELECTED PAPERS

- Max H. Fieg, Patrick J. Fox, **Jinbo Zhang**, Aishik Ghosh, Virat Varada, Daniel Whiteson. “Time-dependent signals of new physics at the LHC.” *arXiv:2605.11071* (2026). Submitted to *Phys. Rev. D*. Report number: FERMILAB-PUB-26-0234-T.

WORK IN PROGRESS

- **Jinbo Zhang**. “A Muon Missing-Momentum Experiment.” *In final preparation for arXiv submission* (2026).
- **Jinbo Zhang**. “Prospects for the measurement of $H \rightarrow WW^*$ in the $q\bar{q} \ell\nu q\bar{q}'$ final state at FCC-ee.” *In Preparation* (2026).
- Rouiela Myles Valencia, Guowen Xing, Anagha Subbaraman, **Jinbo Zhang**, Asantha Cooray, Tom Bakx. “The redshift distribution and physical properties of Herschel-selected ultrared dusty, star-forming galaxies with ALMA Band-3 spectral scans.” *In Preparation* (2026).

OTHER WORK

- Euclid Collaboration: X. Xu, R. Chen, T. Li, A. R. Cooray, . . . , **J. Zhang**, et al. “Euclid Quick Data Release (Q1). AgileLens: A scalable CNN-based pipeline for strong gravitational lens identification.” *arXiv:2604.06648* (2026).
- Christopher M. Gonzalez, Yu-Hsun Chen, Christopher Effarah, **Jinbo Zhang**, Aleksei Anopchenko, Yalun Tang, Kenji Nomura, and Ho Wai (Howard) Lee. “Gate-tunable two-dimensional ITO epsilon-near-zero materials fabricated by liquid metal printing.” *Proc. SPIE PC13579*, PC135791B (2025).
- Ho Wai (Howard) Lee, Quynh Dang, David Dang, Aleksei Anopchenko, Christopher Gonzalez, Stuart Love, Yu-Hsun Chen, Leo Zheng, Meena Salib, **Jinbo Zhang**, et al. “Active and nonlinear epsilon-near-zero photonics.” *Proc. SPIE PC13892* (2026).

POSTERS & TALKS

- **Jinbo Zhang**. “A Muon Missing-Momentum Experiment.” *Final research talk, Caltech SURF program, California Institute of Technology* (Aug. 20, 2026).
- **Jinbo Zhang**. “Time-Dependent Signals of New Physics at the LHC.” *Poster presented at UCI Undergraduate Research Symposium* (May 2026).
- Yu Hsun Chen, **Jinbo Zhang**, et al. “Liquid Metal Printed 2D ITO for Nanophotonic Applications.” *Poster presented at UCI Undergraduate Research Symposium* (May 2025).

TECHNICAL SKILLS

- **Programming & Machine Learning:** Python, C++, PyTorch, XGBoost, NumPy, Pandas, ROOT/PyROOT
- **Physics Software:** Geant4, MadGraph, Pythia, Delphes, FCCAnalyses, pyhf; Slurm-based HPC
- **Languages:** English (Professional Proficiency), Chinese (Native)

TEACHING EXPERIENCE

Learning Assistant: Classical Mechanics / Electromagnetism

Jan. 2025 - Present

University of California, Irvine

- Facilitated discussion sections and laboratory sessions, guiding students through experimental setups, error analysis, and problem-solving strategies for upper-division physics courses.

LEADERSHIP EXPERIENCE

Society of Physics Students at UCI (SPS)

2026 - 2027

Webmaster

Irvine, CA

- Maintain the SPS at UCI website and coordinate with officers to publish events, resources, and outreach information for the undergraduate physics community.

HONORS AND AWARDS

- Caltech SURF Scholarship 2026 - \$8110
- UCI Undergraduate Research Opportunities Program (UROP) Award - 2024, 2025
- Dean's Honor List - All Quarters